

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.ensemble import RandomForestClassifier
```

```
X = [1, 2, 3, 4, 5, None, 7, 8, 9, 10, 11, 12, None, 14, 15, 16, 17, 18, 19, 20]
y_reg = [2, 4, 6, 8, 10, None, 14, 16, 18, 20, 22, 24, None, 28, 30, 32, 34, 36, 38, 40]
y_cls = [0, 0, 0, 0, 0, None, 0, 1, 1, 1, 1, 1, None, 1, 1, 1, 1, 1, 1, 1]
```

```
df = pd.DataFrame({
    'X': X,
    'y_reg': y_reg,
    'y_cls': y_cls
})

print(df)
```

|    | X    | y_reg | y_cls |
|----|------|-------|-------|
| 0  | 1.0  | 2.0   | 0.0   |
| 1  | 2.0  | 4.0   | 0.0   |
| 2  | 3.0  | 6.0   | 0.0   |
| 3  | 4.0  | 8.0   | 0.0   |
| 4  | 5.0  | 10.0  | 0.0   |
| 5  | NaN  | NaN   | NaN   |
| 6  | 7.0  | 14.0  | 0.0   |
| 7  | 8.0  | 16.0  | 1.0   |
| 8  | 9.0  | 18.0  | 1.0   |
| 9  | 10.0 | 20.0  | 1.0   |
| 10 | 11.0 | 22.0  | 1.0   |
| 11 | 12.0 | 24.0  | 1.0   |
| 12 | NaN  | NaN   | NaN   |
| 13 | 14.0 | 28.0  | 1.0   |
| 14 | 15.0 | 30.0  | 1.0   |
| 15 | 16.0 | 32.0  | 1.0   |
| 16 | 17.0 | 34.0  | 1.0   |
| 17 | 18.0 | 36.0  | 1.0   |
| 18 | 19.0 | 38.0  | 1.0   |
| 19 | 20.0 | 40.0  | 1.0   |

```
df = df.dropna()

print(df)
```

|   | X   | y_reg | y_cls |
|---|-----|-------|-------|
| 0 | 1.0 | 2.0   | 0.0   |

```
1    2.0    4.0    0.0
2    3.0    6.0    0.0
3    4.0    8.0    0.0
4    5.0   10.0    0.0
6    7.0   14.0    0.0
7    8.0   16.0    1.0
8    9.0   18.0    1.0
9   10.0   20.0    1.0
10   11.0   22.0    1.0
11   12.0   24.0    1.0
13   14.0   28.0    1.0
14   15.0   30.0    1.0
15   16.0   32.0    1.0
16   17.0   34.0    1.0
17   18.0   36.0    1.0
18   19.0   38.0    1.0
19   20.0   40.0    1.0
```

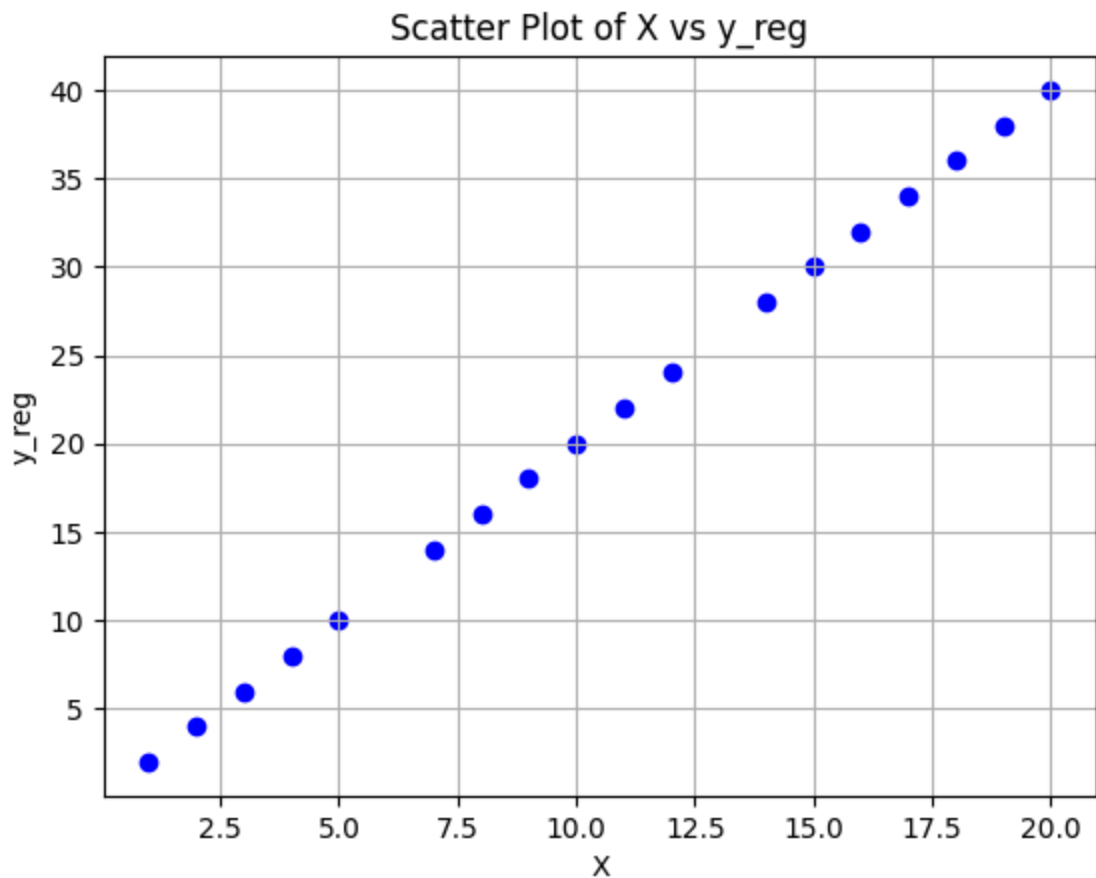
```
X = df[['X']]

y_reg = df['y_reg']

y_cls = df['y_cls']
```

1. Draw a Scatter Plot using Matplotlib for X and y\_reg. Add a title, X-axis label, and Y-axis label

```
plt.scatter(X, y_reg, color="blue")
plt.title("Scatter Plot of X vs y_reg")
plt.xlabel("X")
plt.ylabel("y_reg")
plt.grid(True)
plt.show()
```



2. Create a Linear Regression model and predict the output for X = 7.

```
lr = LinearRegression()
```

```
lr.fit(X, y_reg)
```

▼ LinearRegression ⓘ ?

LinearRegression()

```
pred_lr = lr.predict([[7]])
```

```
print("Linear Regression Prediction:", pred_lr[0])
```

```
Linear Regression Prediction: 14.000000000000002
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning:
  warnings.warn(
```

```
print("Prediction for X = 7:", prediction[0])
```

```
Prediction for X = 7: 14.000000000000002
```

Double-click (or enter) to edit

3. Create a Logistic Regression model and predict the class for  $X = 4$ .

```
log_model = LogisticRegression()
```

```
log_model.fit(X, y_cls)
```

▼ **LogisticRegression** ⓘ ?  
LogisticRegression()

```
pred_log = log_model.predict([[4]])
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning:
warnings.warn(
```

```
print("Logistic Regression Prediction for X = 4 :", pred_log[0])
```

```
Logistic Regression Prediction for X = 4 : 0.0
```

4. Create a KNN Classifier using  $k = 3$  and predict the class for  $X = 5$ .

```
knn = KNeighborsClassifier(n_neighbors=3)
```

```
knn.fit(X, y_cls)
```

▼ **KNeighborsClassifier** ⓘ ?  
KNeighborsClassifier(n\_neighbors=3)

```
pred_knn = knn.predict([[5]])
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning:
warnings.warn(
```

```
print("KNN Prediction for X = 5 :", pred_knn[0])
```

```
KNN Prediction for X = 5 : 0.0
```

5. Create a Random Forest Classifier with 10 trees and predict the class for  $X = 5$ .

```
rf = RandomForestClassifier(n_estimators=10, random_state=42)
```

```
rf.fit(X, y_cls)
```

▼ **RandomForestClassifier** ⓘ ?

```
RandomForestClassifier(n_estimators=10, random_state=42)
```

```
pred_rf = rf.predict([[5]])
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning:
```

```
print("Random Forest Prediction for X = 5 :", pred_rf[0])
```

```
Random Forest Prediction for X = 5 : 0.0
```

6. Print the prediction of each model.

```
print("Linear Regression Prediction for X = 7 :", pred_lr[0])
print("Logistic Regression Prediction for X = 4 :", pred_log[0])
print("KNN Prediction for X = 5 :", pred_knn[0])
print("Random Forest Prediction for X = 5 :", pred_rf[0])
```

```
Linear Regression Prediction for X = 7 : 14.000000000000002
Logistic Regression Prediction for X = 4 : 0.0
KNN Prediction for X = 5 : 0.0
Random Forest Prediction for X = 5 : 0.0
```